MILITARY STANDARD

GAGES, PLUG, PLAIN CYLINDRICAL, NOT GO

TO ALL ACTIVITIES

1. The following pages of this standard have been revised and supersede the pages listed:

New page	*Date*	*Superseded page*	*Date*
1	6 February 1957	1	8 December 1955
3	6 February 1957	3	8 December 1955
5	6 February 1957	5	8 December 1955
6	6 February 1957	6	8 December 1955
7	6 February 1957	7	8 December 1955

2. The following is a cumulative list of earlier changes:

   None.

3. Retain this cover page and insert before table of contents of this standard.

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1. SCOPE

1.1 This standard covers the pertinent design data and stock numbers for American Gage Design Standard, single end, not go plain cylindrical plug gages for the inspection of the maximum limits of internal diameters that range in size from 0.031 to 2.510 inches inclusive in increments of 0.001 inch. Stock numbers are also provided in increments of 0.0005 inch in sizes ranging from 0.031 to 0.8255 inches inclusive for components which have 0.0005 inch total component tolerance. The class of gage required is listed in accordance with the total component tolerance.

1.2 The stock numbers listed in this standard have been approved by the Cataloging Division, Office of the Assistant Secretary of Defense as Federal stock numbers (FSN). See 3.

2. REFERENCED DOCUMENTS

2.1 The following standards, specifications, and publications of latest issue form a part of this standard:

STANDARDS

MIL–STD–10 — Surface Roughness, Waviness, and Lay

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

SPECIFICATIONS

MIL–G–10944 — Gages, Dimensional Control

(Copies of specifications should be obtained from the procuring activity or as directed by the contracting officer.)

OTHER PUBLICATIONS

U. S. Department of Commerce, Commercial Standard CS8, Gage Blanks

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

3. DEFINITIONS

3.1 The Federal stock number (FSN) consists of the applicable 7-digit Federal item identification number designated in table I as "FIIN—Gage Stock No." prefixed by the Federal supply classification (FSC), "5220". The Federal stock number shall be utilized in all instances and specified thus: 4 digits, hyphen, 3 digits, hyphen, 4 digits. (Example: 5220–746–5686.)

4. GENERAL REQUIREMENTS

4.1 Table I. Tabulation of Design and Identification Data.

4.1.1 Stock numbers are listed in ascending numerical sequence in column B.

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4.1.2 Column A lists the proper reference to the applicable figure for each stock number.

4.1.3 Column C lists the maximum component dimension and when applied to the total component tolerance in column D provides the data required for selecting the applicable stock number in column B.

4.1.3.1 When the total component tolerance is not listed, the stock number applicable to the next smaller listed total component tolerance for the maximum component dimension shall be used.

4.1.4 Columns E to G inclusive list the design data for each stock number and, when applied to the figure referenced in column A, provide the data required for manufacture and inspection.

4.1.5 Columns B and C list the identification data for each stock number and, when applied to the figure referenced in column A, provide the information for proper marking of the gages.

4.1.6 Design data, etc. not specifically covered in this standard shall conform to the latest issue of the documents referenced in paragraph 2 above.

5. DETAIL REQUIREMENTS

5.1 Manufacture and inspection.

5.1.1 *Gaging members.*

5.1.1.1 American Gage Design Standard (AGD) gaging members shall be of hardened tool steel or other wear resistant materials suitable for the intended purpose, as defined in specification MIL–G–10944, and shall be identified with the manufacturer's name or trade-mark. (See figs. 2 to 4 incl for hardness requirements.)

5.1.1.2 Not go plain cylindrical plug gages up to and including .510 inch in diameter shall be finished from AGD plain plug gage blanks of the wire type design. Gages above .510 inch up to and including 1.510 inches in diameter shall be finished from AGD not go plain plug gage blanks of the taper lock design with precision centers. Gages above 1.510 inches in diameter shall be finished from AGD not go plain plug gage blanks of the trilock design.

5.1.1.3 The surface roughness value denoted on the gaging surface of figures 2 to 4 inclusive, and listed in column G of table I is the maximum acceptable roughness of these surfaces. These values correspond to the meter indications of surface roughness measuring instruments, which from their method of operation show readings in accordance with the standard values prescribed in Standard MIL–STD–10.

5.1.1.4 Sharp edges shall not be permitted.

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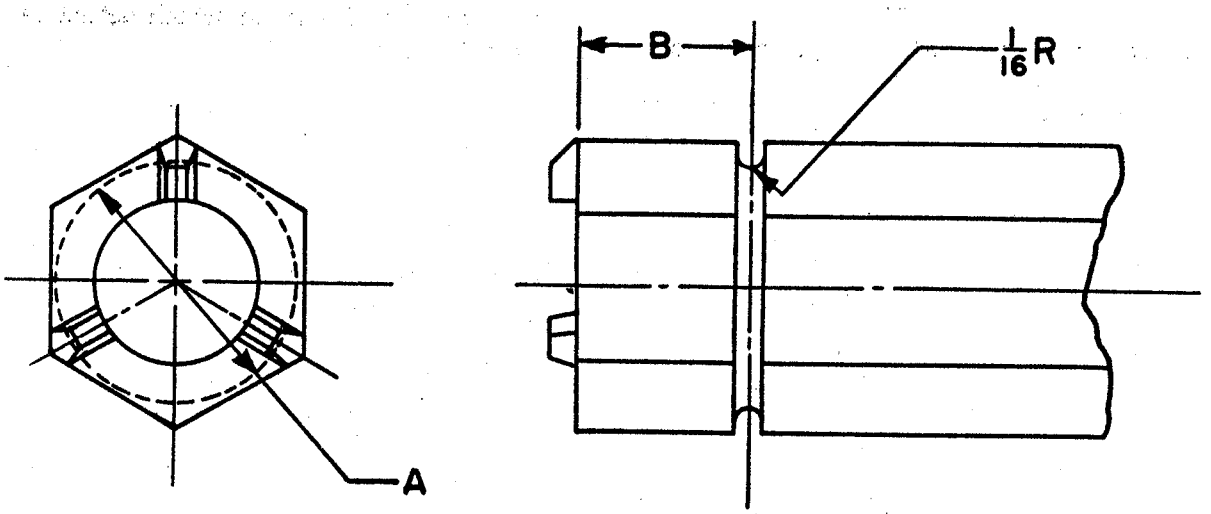
5.1.1.5 Gaging surfaces shall be lapped free of amorphous metal.

5.1.2 *Gage handles.*

5.1.2.1 Gage handles shall be as designated in Commercial Standard CS8. Handle sizes up to and including No. 5 shall be of steel or aluminum alloy. Handle sizes Nos. 5½ and 6 shall be of aluminum alloy. Trilock handles shall be provided with steel screws.

5.1.2.2 All gage handles shall be marked as specified in figures 2, 3, and 4 whichever is applicable. All marking shall be permanent and legible and of suitable height commensurate with the size of the handle.

5.1.2.3 Handle sizes Nos. 5½ and 6 (fig. 4) shall have an annular groove as indicated in figure 1 to identify the gage as a not go gage. However, in lieu of the annular groove, the handles may be finished with red color applied in a durable fashion.



HANDLE NUMBER	A		B
	MIN	MAX	
5½	.975	.980	1
6	1.100	1.105	1

FIGURE 1. *Annular groove.*

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5.2 **Procurement.**

5.2.1 The tabulated design data, manufacturing, and inspection specifications in this standard are applicable for procurement purposes.

5.2.1.1 Procurement documents shall specify the following:

a. Title, number, and date of this standard.

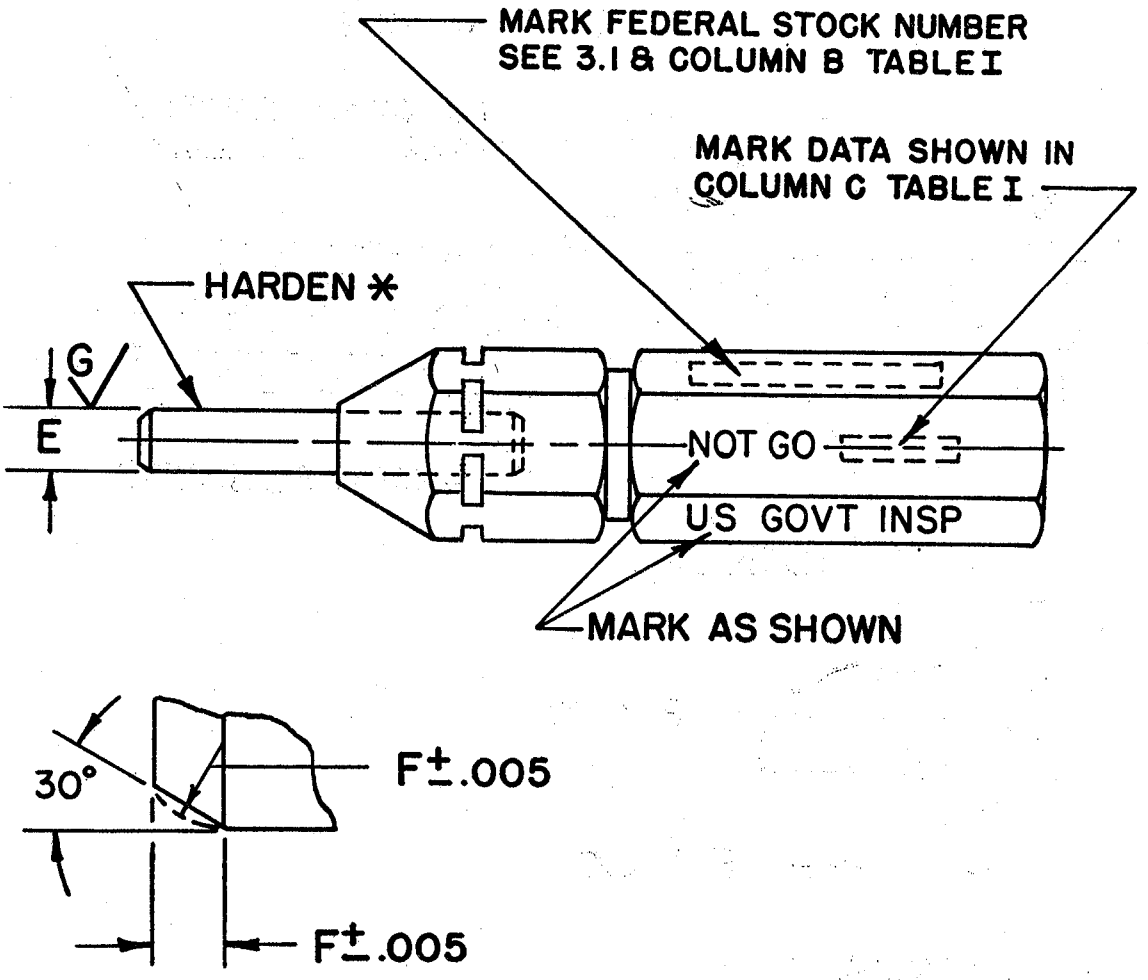
b. Gages required (quantity and stock number).

c. Place of delivery for acceptance inspection or place of delivery if acceptance inspection is to be at contractor's plant.

5.2.1.2 When specified, not go plain plug gages may be procured without handles. Taper lock and trilock gaging members shall be marked with the stock number, the manufacturer's name or trade-mark, and any other pertinent identification data as specified. The wire type gaging member shall be identified with a tag or label marked with the stock number and any other pertinent identification data as specified. The tag or label shall be such that it can be removed without damage to the member.

5.2.1.3 *Caution.* Although all items listed in this standard are included in the Federal Catalog system as items of supply, it is not intended that all items be procured and stocked. Procurement and supply shall be confined to those items for which actual requirements exist.

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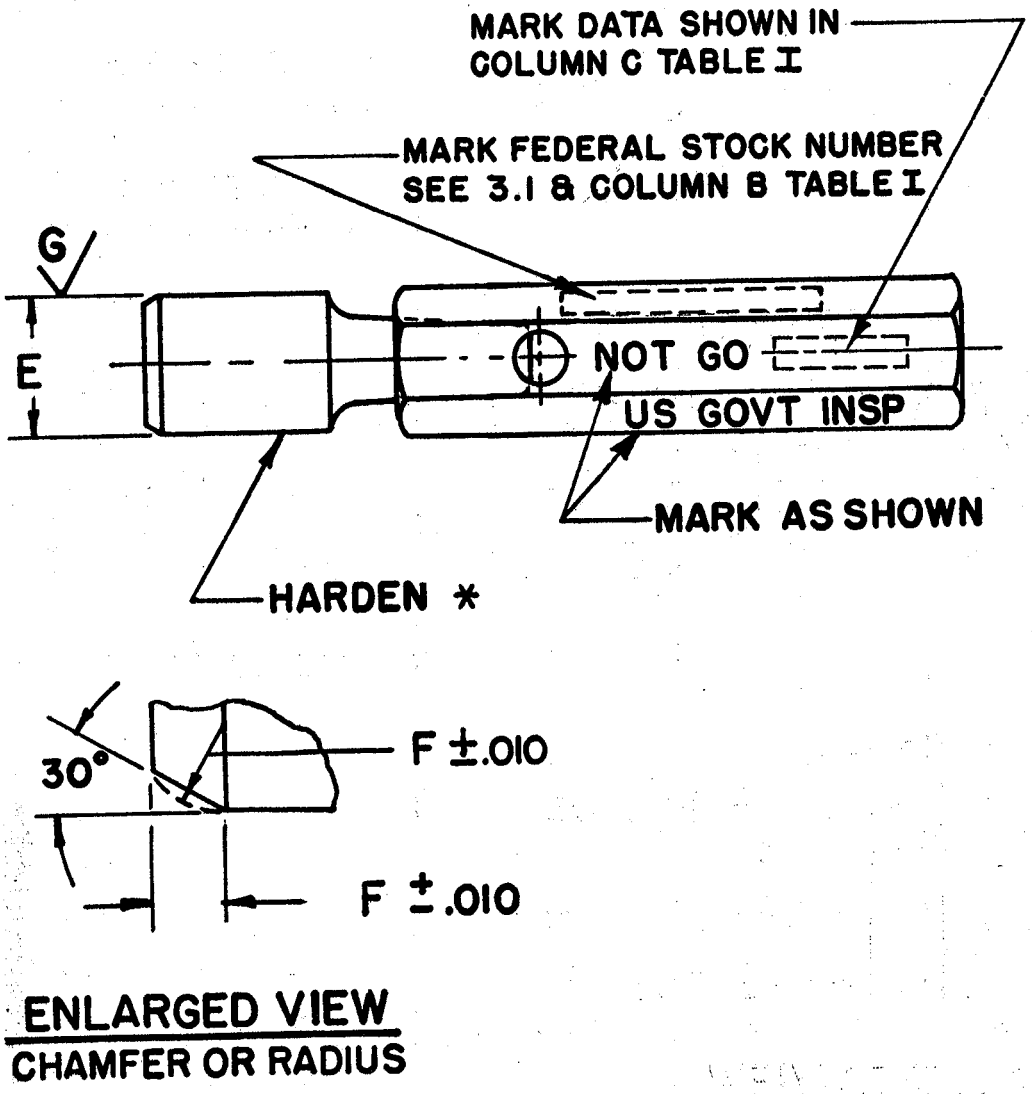
*HARDEN:ROCKWELL OR EQUIVALENT

C55 TO C60 FOR DIA UP TO AND INCL .10
C60 TO C63 FOR DIA ABOVE .10 TO AND INCL.20
C63 TO C66 FOR DIA ABOVE .20

FIGURE 2. *Not go plain plug gage (wire type).*

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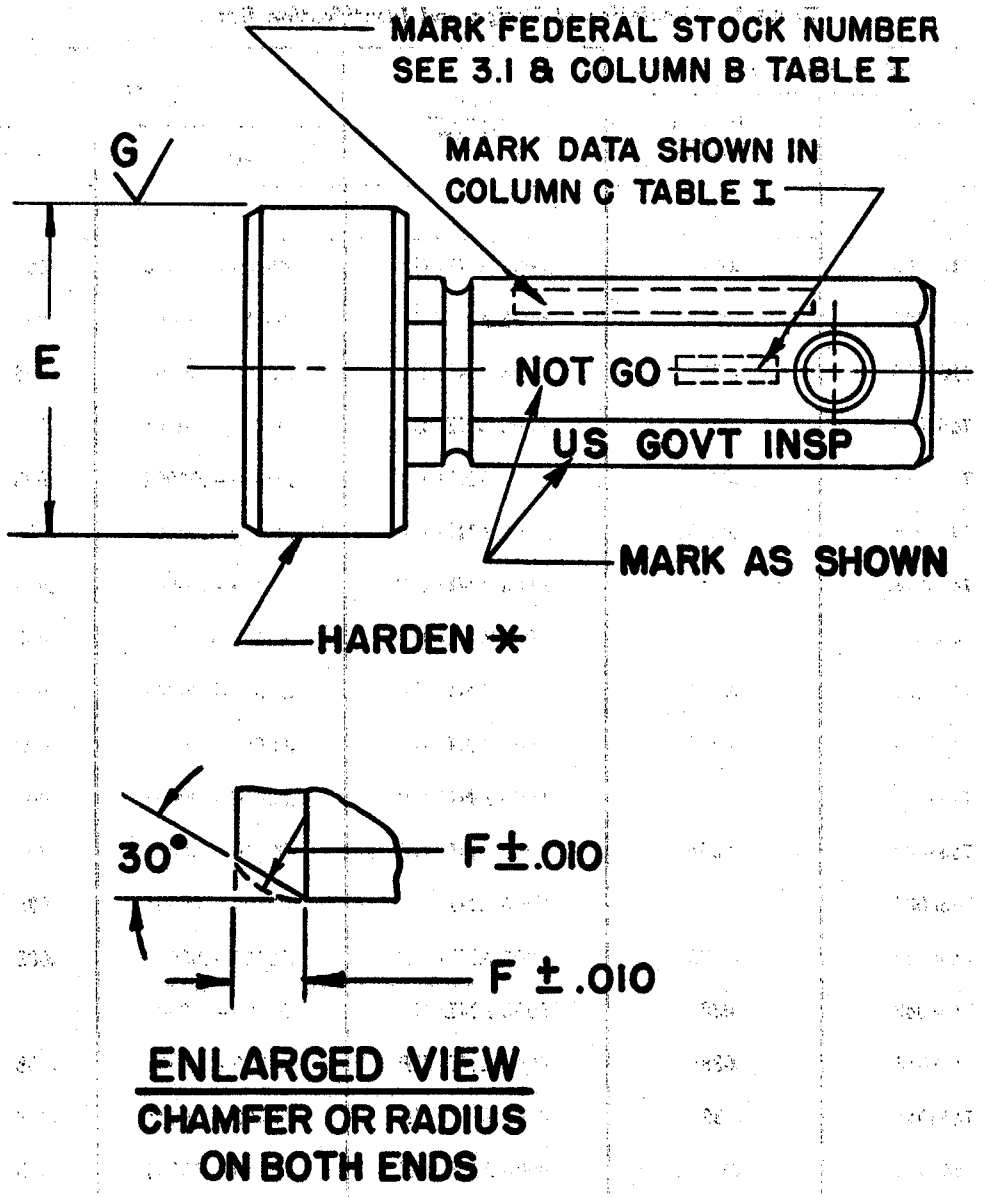


* HARDEN: ROCKWELL OR EQUIVALENT
C63 TO C66

FIGURE 3. *Not go plain plug gage (taper lock type).*

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* HARDEN: ROCKWELL OR EQUIVALENT
C 63 TO C66

FIGURE 4. *Not go plain plug gage (tri-lock type).*

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TABLE I. *Tabulation of Design and Identification Data.*

Fig. No.	Marking on gage handle		Total Component tolerance	Plug gage dimension		Finish
	FIIN Gage Stock No.	Maximum component dimension		Diameter	Chanfer	
A	B	C	D	E	F	G
2	7458000	.031	.0005 AND UP	.03100 — .00004	.008	4
2	7458004	.0315	.0005 AND UP	.03150 — .00004	.008	4
2	7458006	.032	.0005 AND UP	.03200 — .00004	.008	4
2	7458010	.0325	.0005 AND UP	.03250 — .00004	.008	4
2	7458012	.033	.0005 AND UP	.03300 — .00004	.008	4
2	7458016	.0335	.0005 AND UP	.03350 — .00004	.008	4
2	7458018	.034	.0005 AND UP	.03400 — .00004	.008	4
2	7458022	.0345	.0005 AND UP	.03450 — .00004	.008	4
2	7458024	.035	.0005 AND UP	.03500 — .00004	.008	4
2	7458028	.0355	.0005 AND UP	.03550 — .00004	.008	4
2	7458030	.036	.0005 AND UP	.03600 — .00004	.008	4
2	7458034	.0365	.0005 AND UP	.03650 — .00004	.008	4
2	7458036	.037	.0005 AND UP	.03700 — .00004	.008	4
2	7458040	.0375	.0005 AND UP	.03750 — .00004	.008	4
2	7458042	.038	.0005 AND UP	.03800 — .00004	.008	4
2	7458046	.0385	.0005 AND UP	.03850 — .00004	.008	4
2	7458048	.039	.0005 AND UP	.03900 — .00004	.008	4
2	7458052	.0395	.0005 AND UP	.03950 — .00004	.008	4

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